

The dominated-coordinate transfer lemma

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Abstract

Second step of the mixed-ratio general- d monomial programme (Astra #15, route A). For $q > \lambda > 0$ and a bounded measurable f on $(0, \infty)$ with $f(M)/(M^{-\lambda}(\log M)^r) \rightarrow C$,

$$\frac{\int_0^1 t^{q-1} f(Nt) dt}{N^{-\lambda}(\log N)^r} \longrightarrow \frac{C}{q - \lambda}$$

(tendsto_weighted_scale_div): integrating against a dominated coordinate weight preserves the exponent and the logarithmic degree and multiplies the constant by $1/(q - \lambda)$. Zero sorry/axiom.

1 The things formalised

```
theorem integral_Ioc_rpow_gap (l q) (hq : l < q) :
  t in Ioc 0 1, t ^ (q - l - 1) = 1 / (q - l)
theorem integrableOn_Ioc_rpow_gap (l q) (hq : l < q) :
  IntegrableOn (fun t : => t ^ (q - l - 1)) (Ioc 0 1)
theorem tendsto_weighted_scale_div (f) (l q) (r) (C B) (hl : 0 < l) (hq : l < q)
  (hf : Measurable f) (hB : M, 0 < M → |f M| ≤ B)
  (hlim : Tendsto (fun M => f M / (M ^ (-l) * log M ^ r)) atTop (C)) :
  Tendsto (fun N => ( t in Ioc 0 1, t ^ (q - l) * f (N * t)) / (N ^ (-l) * log N ^ r))
    atTop (C / (q - l)))
```

2 Mathematics

Dominated convergence on $(0, 1]$ along $N \rightarrow \infty$: the integrand $t^{q-1}f(Nt)/(N^{-\lambda}(\log N)^r)$ converges pointwise to $Ct^{q-\lambda-1}$ (unit 177's scaled limit) and is dominated for $N \geq e$ by $K2^r t^{q-\lambda-1}$ (unit 177's scaled envelope), which is integrable because $q > \lambda$; the limit integral is $C/(q - \lambda)$.

3 Paper-facing meaning

This is the analytic content of the paper's residue factor $1/(h_i + \gamma_i + 1 - 2k_i\lambda)$ for a coordinate $i \notin J$ not realising the minimal ratio (eq. `a_minus_m_explicit`, $b = 1$): after the substitution $t_i = u_i^{2k_i}$ it is $\frac{1}{2k_i} \cdot \frac{1}{\lambda_i - \lambda}$. The remaining units build the real weighted box integral, its permutation invariance and the induction that removes nonminimal coordinates one at a time.

4 Lean notes

- `integral_Ioc_rpow_sub_one` already existed (unit on iterated primitives, with a variable upper limit x); a same-named declaration in the namespace is a hard error, so the new helper is `integral_Ioc_rpow_gap`. Grep the namespace before naming helpers.
- `tendsto_integral_filter_of_dominated_convergence` with an eventual bound: the measurability, bound and limit goals are all discharged on `volume.restrict (Ioc 0 1)` via `ae_restrict_iff'`; the bound only holds for $N \geq e$, which the `form` allows.
- `integral_div` converts the DCT conclusion about $\int F_N$ into the ratio form.